LUC@BJTU Weihai Campus

Fundamentals of Communication Systems (CNSCC160) Tutorial 2

Knowledge

1. Differentiate between an analogue and a digital electromagnetic signal.
2. What are three important characteristics of a periodic signal?
3. What is the relationship between a signal's spectrum and its bandwidth?
4. What is attenuation?
5. Define channel capacity.
6. What key factors affect channel capacity?
7. Differentiate between guided media and unguided media.
8. What are some major advantages and disadvantages of microwave transmission?

9、$SNR = \frac{10\times10^{-3}}{1\times10^{-6}} = 10^4$

$SNR_{dB} = 10\cdot\lg SNR = 40\ dB$

$10 = 10\cdot\lg\frac{P_{out}}{P_{in}} = -4.77dB$

$= 10\cdot\lg\frac{P_1}{\frac{1}{3}P_1}$

$= -10\cdot\lg 3$

Understanding

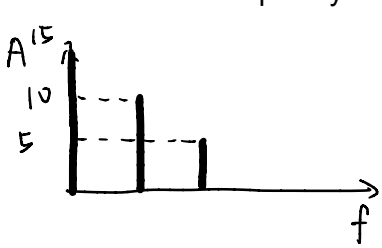
9. The power of a signal is 10 mW and the power of the noise is 1 μW. What are the values of SNR and SNR_{dB}?
10. Suppose a signal travels through a transmission medium and its power is reduced to one-third. This means that P2 = 0.333*P1. Find the attenuation (loss of power) for this case (measured in dB).
11. A signal travels through an amplifier, and its power is increased 5 times. This means that P2 = 5*P1. Find the amplification (gain of power in dB) for this case.

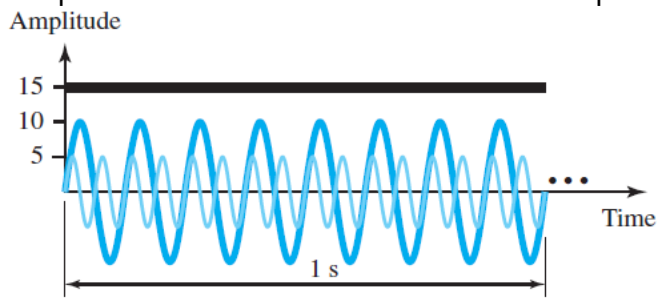
$11 = 10\cdot\lg\frac{5P_1}{P_1}$

$= 10\cdot\lg 5$

$= 6.98\ dB$ 加单位

12. The below figure shows three sine waves, each with different amplitude and frequency. Please represent these three sine waves in the frequency domain.





a. Time-domain representation of three sine waves with frequencies 0, 8, and 16

$N = 2B\log_2 L$

$N = S\cdot r$

$S = 2B$

↓ max S

13. What is the channel capacity for a teleprinter channel with a 300-Hz bandwidth and a signal-to-noise ratio of 3 dB?

$3 = 10\cdot\lg SNR$

$\lg SNR = 0.3$

$SNR = 10^{0.3} = 1.9952$

$C = B\log_2(1+SNR)$

$= 300\times\log_2(1+1.9952)$

$= 474.7957\ bps$

☆ $B = (1+\alpha)\cdot S$

只适用于调制

$N = 2B \cdot \log_2 L$ $\quad 9600 \cdot 2B_{min} \cdot 4, \quad B_{min} = 1200\,Hz$

$9600 \cdot 2B_{min} \cdot 8, \quad B_{min} = 600\,Hz$

14. A digital signalling system is required to operate at 9600 bps.

    a) If a signal element encodes a 4-bit word, what is the minimum required bandwidth of the channel?
    b) Repeat part (a) for the case of 8-bit words.

$N = 2B \cdot \log_2 L \qquad C = B\log_2(1+SNR)$

15. Study the works of Shannon and Nyquist on channel capacity. Each places an upper limit on the bit rate of a channel based on two different approaches. How are the two related?

Limited by bandwidth and L, Limited by Bandwidth and SNR

16. Given a channel with an intended capacity of 20 Mbps, the bandwidth of the channel is 3 MHz. What signal-to-noise ratio is required to achieve this capacity?

17. Consider a series in which the input is at a power level of 4 mW, the first element is a transmission line with a 12-dB loss (-12 dB gain), the second element is an amplifier with a 35-dB gain, and the third element is a transmission line with a 10-dB loss.
    Calculate the net gain and output power.

16、 $20\times10^6 = 3\times10^6 \cdot \log_2(1+SNR)$

$2^{\frac{20}{3}} = 1+SNR$

$SNR = 100.59$

17、 net gain

$= -12+35-10$

$= 13\,dB$

$13 = 10 \cdot \lg \frac{P}{4\times10^{-3}}$

$\frac{P}{4\times10^{-3}} = 10^{1.3}$

$= 0.0798\,W$